MUHAMMAD SAAD AYAZ

AI/ML Engineer | Agentic AI, LLMs & RAG Systems

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SUMMARY

AI/ML Engineer with ~2 years building and deploying production AI systems across LLM applications, agentic pipelines, and computer vision. Hands-on with multi-agent orchestration (LangGraph), RAG, FastAPI, and full-stack deployment, owning features end to end from model integration to production.

TECHNICAL SKILLS

AI / ML & LLMs: Python, LLMs (OpenAI, Claude, Gemini), RAG, LangGraph, LangChain, NLP, Computer Vision, PyTorch, TensorFlow, Generative AI, Agentic AI Specialist, Claude and OpenAI Specialist.

Backend & Infra: FastAPI, RestAPI, Django, django-tenants, RabbitMQ, Redis, PostgreSQL, Supabase, Playwright, Ollama, ComfyUI, Docker, Celery, Containerization, Production Deployment.

Frontend & Mobile: React, Next.js, TypeScript, Vite, Node.js, Capacitor (Android/iOS), Git & GitHub

Languages: Python, TypeScript/JavaScript, Java, C#, SQL, Node JS

WORK EXPERIENCE

AI/ML Engineer | National Center for Cyber Security (NCCS), NED University | 06/2026 – Present

- Building Attack-Eye, a multi-tenant security scanning platform (Django, django-tenants, Celery, RabbitMQ, PostgreSQL, Redis) with Ollama for local LLM inference and Uptime Kuma monitoring across 24 service checks.
- Delivered an end-to-end on-premise, white-labeled deployment for LuckyOne (Docker, Nginx reverse proxy with TLS, multi-tenant hostname routing) and diagnosed & production blockers spanning static-asset delivery, message-broker compatibility, and build dependencies.
- Shipped post-launch dashboard features (real-time Alerts panel, per-tenant Team Activity view) with schema migrations across every tenant.

Software Engineer | KMNR Engineering Pvt Ltd | 06/2025 – 06/2026

- Built an internal Vacation Request System (Blazor, .NET, SQL) tracking leave balances across four leave types, and Skills Hub, a skill-tracking system giving managers visibility into team progress.

AI/ML Trainee Engineer | Enterprise64 | 11/2024 – 02/2025

- Developed a real-time facialrecognition attendance system for an HR platform using CCTV feeds (MediaPipe, facerecognition, dlib).
- Built a voice-based PTO request system using LLMs (Gemini) and OpenAI Whisper V3 with automatic leave-type suggestion.

Machine Learning Intern | Softech Worldwide | 09/2023 – 11/2023

- Developed and fine-tuned models for object/vehicle detection and a facialrecognition system for detection and verification.

PROJECTS

JARVIS — Multi-Agent AI Growth & Content Platform **| Python ▪ FastAPI ▪ PostgreSQL ▪ Next.js ▪ Docker**
(solo-built, zero budget)

- **Architected a voice-controlled, 9-agent AI operating system covering content creation, AI-search visibility scoring, outbound sales automation, and AI video/ai generation, driven by one natural-language command layer and a provider-agnostic orchestration engine.**
- **Built a voice-first control console (Next.js, Web Speech API, WebSocket) with live agent task status, plus an autonomous background worker so scheduled tasks keep running independent of the UI.**
- **Engineered an LLM abstraction layer with hot-swapable providers (Gemini, OpenAI, Claude) and a cascading image-generation fallback chain (OpenRouter → OpenAI → native → Pollinations) to balance cost and reliability.**
- **Built GEO (Generative Engine Optimization), an original pipeline scoring brand visibility in AI answer engines (ChatGPT, Perplexity, Google AI Overviews) against mainst competitors on a 0–100 presence/citation/share-of-voice scale.**
- **Developed an AI video pipeline (script → free TTS → B-roll via local ComfyUI/Hugging Face → ffmpeg assembly with captions → optional YouTube upload) entirely on free/self-hosted infra, plus an outbound sales agent.**
- **Enforced approval guardrails on every publishing/outreach surface (no post, upload, or send without explicit approval); secured the control plane with token auth; deployed via Docker Compose and Cloudflare Tunnel.**

github.com/saadayazzzz/Multi-AI-Agents

AI-Powered Sales & Automation Platform | Python | FastAPI ▪ PostgreSQL ▪ Next.js ▪ Playwright

- **Designed and built a multi-agent orchestration system that routes natural-language tasks to specialized agents (sales, content, GEO/visibility, video) via an LLM-driven tool-calling orchestrator (Claude, GPT, Gemini).**
- **Built a self-hosted B2B lead-generation engine: signal-based LinkedIn sourcing (recent activity, job changes, funding events), AI lead scoring, and automated pitch/outreach drafting, with Notion CRM sync generating a pitch email and LinkedIn message per lead.**
- **Integrated Playwright-based LinkedIn automation (Node.js subprocess) with per-seat daily rate limits and a manual-approval gate on every outbound send, Postgres-backed async task queue decouples the API from long-running agent work.**
- **Developed a real-time operations dashboard (Next.js, TypeScript, WebSocket) for live agent activity task queues, and lead pipelines; retrofitted bearer token auth across a previously open API surface and deployed via Cloudflare Tunnel with CORS-hardened access.**

Real-Time Voice AI Agent | OpenAI Realtime API ▪ FastAPI ▪ Vapi ▪ Docker

- **Full-duplex voice agent on the OpenAI Realtime API (STT, inference, synthesis in one connection): transport agnostic browser/phone (Vapi) bridge, server-side VAD barge-in handling, idrempotent transcript writes, 23-case test suite.** _github.com/saadayazzzz/AI-Voice-Agent

Multi-Agent Job-Matching Pipeline | LangGraph ▪ FastAPI ▪ React

- **Two-agent LangGraph pipeline (Scorer + Drafter), conditional state graph, FastAPI + React.**

github.com/saadayazzzz/Multi-Agent-Job-Matching-Pipeline

SSU FitTest — Cross-Platform Fitness Testing App | React ▪ Capacitor ▪ Supabase ▪ MediaPipe

- **Web + Android app: offline on-device rep counting (MediaPipe Pose), GPS run tracking, offline-first Supabase sync (RLS), SVG analytics.**

EDUCATION

BS, Computer Science | PAF-KIET (Karachi Institute of Economics & Technology) | 01/2020 – 08/2024

Intermediate, Pre-Engineering | Delhi Science & Commerce College | 08/2017 – 05/2019

CERTIFICATIONS

ML Specialization - AI Python for Everybody (in progress) ▪ Git & GitHub ▪ Cloud Computing ▪ DevOps — DeepLearning-AI, U. Michigan, Coursera